

# Report

Title: Configuring the NGINX on AWS EC2

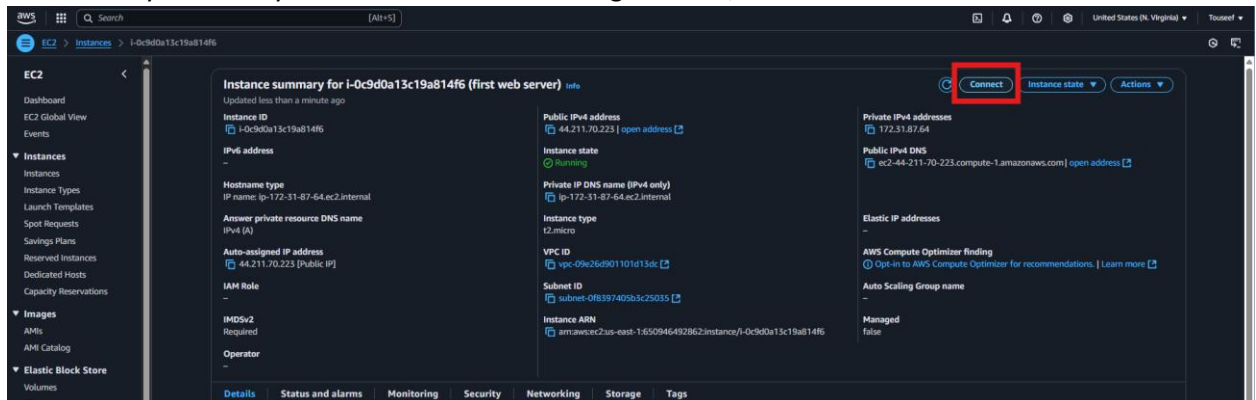
## Setup

Create a fresh EC2 instance with following configs;

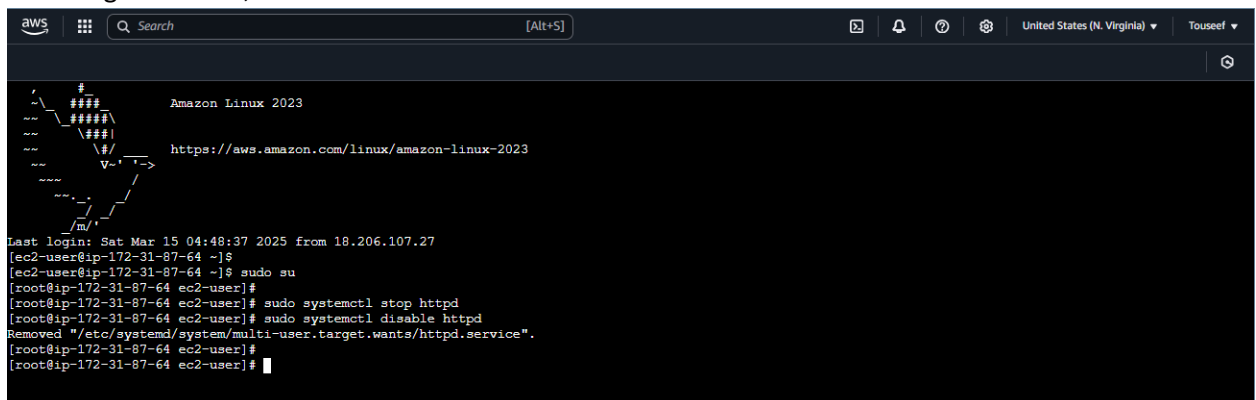
- No Key Value Pair Encryption
- Allow http/https Traffic
- Everything else to default values

## Procedure

1. Connect to your freshly created AWS instance using browser;



2. Switch to root user, if Apache Server already configured on the machine, stop the Apache Server and disable the Apache Server service (so it does not auto start when EC2 restarts) with following command;



- Update the machine packages with following command;

```
aws [Alt+S] United States (N. Virginia) Touseef

[root@ip-172-31-87-64 ec2-user]#
[root@ip-172-31-87-64 ec2-user]#
[root@ip-172-31-87-64 ec2-user]#
[root@ip-172-31-87-64 ec2-user]# yum update -y
Amazon Linux 2023 Kernel Livepatch repository 112 kB/s | 14 kB 00:00
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-172-31-87-64 ec2-user]#
```

- Install the Nginx server using the following command;

```
aws [Alt+S] United States (N. Virginia) Touseef

[root@ip-172-31-87-64 ec2-user]#
[root@ip-172-31-87-64 ec2-user]#
[root@ip-172-31-87-64 ec2-user]# sudo yum install nginx -y
Last metadata expiration check: 0:34:10 ago on Sat Mar 15 04:49:59 2025.
Dependencies resolved.

Package Architecture Version Repository Size
Installing:
nginx x86_64 1:1.26.3-1.amzn2023.0.1 amazonlinux 33 k
Installing dependencies:
gperftools-libs x86_64 2.9.1-1.amzn2023.0.3 amazonlinux 308 k
libunwind x86_64 1.4.0-5.amzn2023.0.2 amazonlinux 66 k
nginx-core x86_64 1:1.26.3-1.amzn2023.0.1 amazonlinux 670 k
nginxfilesystem noarch 1:1.26.3-1.amzn2023.0.1 amazonlinux 9.6 k
nginx-mimetypes noarch 2.1.49-3.amzn2023.0.3 amazonlinux 21 k

Transaction Summary
Install 6 Packages

Total download size: 1.1 M
Installed size: 3.6 M
Downloading Packages:
(1/6): libunwind-1.4.0-5.amzn2023.0.2.x86_64.rpm 1.5 MB/s | 66 kB 00:00
(2/6): nginx-1.26.3-1.amzn2023.0.1.x86_64.rpm 698 kB/s | 33 kB 00:00
(3/6): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64.rpm 5.1 MB/s | 308 kB 00:00
(4/6): nginxfilesystem-1.26.3-1.amzn2023.0.1.noarch.rpm 420 kB/s | 9.6 kB 00:00
(5/6): nginx-core-1.26.3-1.amzn2023.0.1.x86_64.rpm 18 MB/s | 670 kB 00:00
(6/6): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch.rpm 824 kB/s | 21 kB 00:00

Total 9.2 MB/s | 1.1 MB 00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing :
Running scriptlet: nginxfilesystem-1:1.26.3-1.amzn2023.0.1.noarch 1/1
Installing : nginxfilesystem-1:1.26.3-1.amzn2023.0.1.noarch 1/6
Installing : nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch 2/6
Installing : libunwind-1.4.0-5.amzn2023.0.2.x86_64 3/6
Installing : gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64 4/6
Installing : nginx-core-1:1.26.3-1.amzn2023.0.1.x86_64 5/6
Installing : nginx-1:1.26.3-1.amzn2023.0.1.x86_64 6/6
Running scriptlet: nginx-1:1.26.3-1.amzn2023.0.1.x86_64 6/6
Verifying : gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64 1/6

I-0c9d0a13c19a814f6 (first web server)
PublicIPs: 44.204.174.36 PrivateIPs: 172.31.87.64
```

5. Start the installed Nginx server using the following command;

```
aws [Search] [Alt+S] United States (N. Virginia) Touseef
[root@ip-172-31-87-64 ec2-user]#
[root@ip-172-31-87-64 ec2-user]#
[root@ip-172-31-87-64 ec2-user]# sudo systemctl start nginx
[root@ip-172-31-87-64 ec2-user]#
```

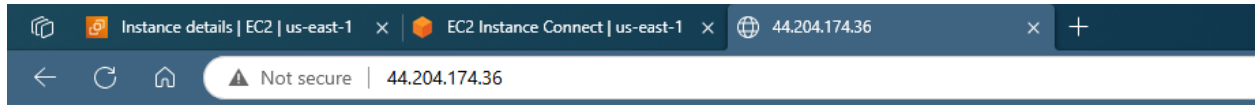
6. Also enable the installed Nginx server service so that it can auto start when the EC2 instance restarts. Use the following command;

```
aws [Search] [Alt+S] United States (N. Virginia) Touseef
[root@ip-172-31-87-64 ec2-user]#
[root@ip-172-31-87-64 ec2-user]#
[root@ip-172-31-87-64 ec2-user]# sudo systemctl enable nginx
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service -> /usr/lib/systemd/system/nginx.service.
[root@ip-172-31-87-64 ec2-user]#
```

7. Create a custom index.html file at /usr/share/nginx/html/index.html path of your machine which would be automatically returned by the web server upon client web request.

```
aws [Search] [Alt+S] United States (N. Virginia) Touseef
[root@ip-172-31-87-64 ec2-user]#
[root@ip-172-31-87-64 ec2-user]#
[root@ip-172-31-87-64 ec2-user]# echo "My amazing Nginx Server!" > /usr/share/nginx/html/index.html
[root@ip-172-31-87-64 ec2-user]#
```

- Now copy the public IP of your EC2 machine and send the http request from your browser. You should be able to see content of your index.html file in the browser body.



My amazing Nginx Server!